

Perfect Mode Hal: Aggressive, Change-Aware Opponent Exploitation in Exact DTH

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Abstract

Perfect Mode (PM) Hal is a causal meta-policy for repeated, pure Drop the Handkerchief (DTH). One certified tablebase remains the sole authority for each continuation-adjusted 60×60 matrix. Around it, PM combines Adaptive Hal’s Bayesian counts, Perfect Hal’s pattern experts, online change-point and outcome models, and a compatible Aggro Hal recurrent network. Fixed Share learns role-specific source and controller weights. Four modes move from an exact shield to hard forecast responses, while every selected policy receives a fresh local worst-case-loss measurement. An earlier 24-identity development run scored 44 wins in 48 games, but its protocol lacked durable registration and its evaluator trusted self-reported risk. It has no promotion authority. This note instead registers a 56-identity confirmation protocol with independent matrix recomputation, a no-Aggro ablation, and family and seat gates. At the protocol commit that confirmation had not been executed. The one-shot run scored PM at .8705 (97–14–1), behind Perfect at .9554; the promotion gate failed despite zero independent local-risk audit violations. PM is an aggressive candidate, not human mind reading, dominance, or a whole-game safety theorem.

1 Scope and design philosophy

At public state x , the exact DTH agent supplies matrix $M(x)$, equilibrium policies (x^*, y^*) , and a saddle-gap certificate at most 10^{-6} . All actions are literal seconds $1, \dots, 60$. PM never estimates continuation values and never sees an unrevealed simultaneous action.

The intended character is “overbearing after evidence, composed after surprise.” Perfect contributes hard exploitation; Adaptive contributes uncertainty and a constrained response frontier; Aggro contributes recurrent features and a learned residual policy. The story evidence ledger motivates competing personae, opponent mental models, and abrupt memory discontinuities. PM translates that fiction into testable engineering—model competition, role separation, recency, and change recovery—rather than treating a person as one fixed psychological type. The final narrative ideal, to improve and adapt without looking back, becomes persistent memory with rapid de-weighting and recovery after changed play.

2 Causal opponent model

For each opponent role separately, PM maintains six mandatory forecast sources: uniform recovery, the opponent’s exact equilibrium policy, a decayed Dirichlet model, Perfect’s online expert mixture, categorical Bayesian online change-point detection (BOCPD), and an outcome-conditioned model. A seventh source is Aggro’s GRU forecast when a strict compatible checkpoint is present. After action a_t is revealed, source k receives prequential log score and Fixed Share updates

$$\begin{aligned}\tilde{w}_{k,t+1} &\propto w_{k,t} \exp(\eta \log q_{k,t}(a_t)), \\ w_{k,t+1} &= (1 - \alpha)\tilde{w}_{k,t+1} + \frac{\alpha}{K - 1} (1 - \tilde{w}_{k,t+1}),\end{aligned}$$

with $\alpha = .04$. Thus every source retains recovery mass when an opponent switches. BOCPD carries a truncated Dirichlet–multinomial run-length posterior with hazard .08. The fused forecast is $\hat{q}_t = \sum_k w_{k,t} q_{k,t}$.

Aggression confidence is the declared heuristic

$$c_t = c_{\max} \frac{n_{\text{eff}}}{4 + n_{\text{eff}}} (1 - P_t(\text{change}))(1 - \text{JS}_t)s_t,$$

where JS_t is normalized source disagreement and $s_t = \exp[-\max(0, \overline{\text{NLL}}_t - \log 60)]$. Only the evidence fraction has a data-biased-response interpretation; the product is a project controller, not a published theorem.

3 Aggression with measured risk

For Dropper, the certified ϵ candidate solves

$$\max_{x \in \Delta_{60}} x^\top M \hat{q}_t \quad \text{s.t.} \quad (M^\top x)_j \geq L - \epsilon \quad \forall j,$$

where $L = \min_j (M^\top x^*)_j$ is freshly recomputed. Checker uses the symmetric upper constraint. The bank also contains exact equilibrium, hard responses to fused and Perfect forecasts, and Aggro’s direct policy. For a direct Dropper policy x , actual local loss is $\max(0, L - \min_j (M^\top x)_j)$; every direct and frontier candidate is independently checked against the current matrix.

One common bank of heuristic ensemble perturbations must give positive expected improvement and at least .55 positive-draw support. These draws are not a calibrated posterior, so the support fraction is not an improvement probability. Cold start or a change shock enters *shield*,

where only the exact face is admissible. The registered v3 caps are *probe* .05, *press* .50, and *dominate* 2.0, inside a 12.0 per-game charge budget. Controller weights receive full-information counterfactual payoff feedback after reveal. This is deliberately aggressive: the budget is an explicit risk appetite, not a guarantee that losses add across a DTH game. Ganzfried–Sandholm’s cumulative ledger concerns repetitions of one stage game; DTH’s matrices change with the state and already contain overlapping continuation values.

4 Development audit and registered confirmation

The v1 and v2 files were run before this branch registered them in Git; mutable status text cannot establish preregistration. They remain development evidence. V2 used 24 identities across six families, common seeded seat pairs, and 10,000 identity-clustered bootstrap replicates. Stopped games scored .5.

Table 1: Development v2 and registered v3 outcomes.

Controller	V2 score	V3 score	V3 W–L–S
PM Hal	.9167	.8705	97–14–1
Perfect	.8958	.9554	107–5–0
PM no Aggro	—	.8259	92–19–1
Aggro	.6458	.7232	81–31–0
Adaptive	.6563	.7188	80–31–1
Exact	.5417	.4955	54–55–3

V2’s nominal every-component gate failed. Its controller-reported risk was not independently reconstructed, so it cannot promote PM. Fused expected NLL was 3.481 [2.904, 3.978], but change detection found only 10 of 86 truth shifts. These mixed results motivated the registered repair.

Clean-Git v3 hash-binds the controller, recurrent checkpoint, and tablebase, records its commit, and uses 56 fresh identities over all 14 families (112 games per controller). It compares PM with Exact, Perfect, Adaptive, Aggro, and PM without Aggro. Promotion requires zero independent risk violations and nonnegative aggregate, both-seat, and every-family comparisons. Every selected policy’s loss, charge contract, mode cap, cumulative charge, and game budget is recomputed from its continuation-adjusted matrix.

The registered protocol ran once from commit `b76147f`. PM-minus-baseline paired score differences were $-.0848$ $[-.1518, -.0223]$ against Perfect, $+.0446$ $[.0089, .0893]$ against PM without Aggro, and $+.3750$ $[.2723, .4777]$ against Exact. The gate failed the every-component, both-seat, and every-family conditions; Perfect was the strongest component. The independent audit found zero violations over 788 decisions and 112 games, with maximum measured local loss 1.5373. Fused expected NLL was 3.5524 versus uniform 4.0943, but the change detector found only 35 of

168 truth shifts and produced 23 false-alarm onsets. The result supports aggregate value from enabling the Aggro component in this protocol; it does not isolate recurrent memory as the cause. The hash-bound compact record is `pm_hal_confirmation_v3.json`.

5 Human-play interpretation and limits

The psychology is operational: reinforcement, belief learning, response to one’s own revealed action, recency, periodicity, and change are competing causal hypotheses. A human can counter-adapt, deceive, or change strategy; PM must preserve recovery mass and retreat on surprise. It should never label a participant’s personality or claim privileged access to intent.

A human study must keep opponent identity across games, log the sequence forecast before each reveal, score realized NLL/Brier without simulator-truth distributions, and cluster outcomes by participant. Change delay and false alarms, role slices, calibration, and game score are co-primary diagnostics. Bounded recursive reasoning is one possible population archetype, not a label the controller may assign permanently to a participant. The present measured evidence comprises development evidence from 24 identities and one registered synthetic confirmation over 56 fresh identities and known families. The recurrent checkpoint saw 2,653 training decisions. Its enabled component improved aggregate score over the no-Aggro ablation, while a causal recurrent-memory advantage remains open. PM is feature-complete for human play as an aggressive candidate. Synthetic promotion failed; any future protocol requires a new version and fresh identities, followed by participant-level human evidence.

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